Empirical and Theoretical Probability

Contents

1. Overview . . . 2
2. What a probability measures . . . 3
3. Empirical probability . . . 3
    3.1 A bag of marbles . . . 4
    3.2 Titanic passengers . . . 4
4. Theoretical probability . . . 5
5. From empirical to theoretical . . . 8
    5.1 Which one to use . . . 9
6. The rules every probability follows . . . 9
    6.1 The axioms of probability . . . 10
    6.2 The complement rule . . . 10
    6.3 The general addition rule . . . 11
7. Summary . . . 13
8. Sources . . . 13
9. Key terms . . . 14

Where this fits: Pipeline map step 0 (Foundations). See the Course map.

0 Foundations	1 Frame	2 Get data	3 Understand	4 Clean	5 Features	6 Reduce
7 Split	8 Model	9 Evaluate	10 Tune	11 Deploy	12 Test	13 Monitor

- **Builds on:** Events and sample spaces.
- **Leads to:** Random variables.

1. Overview

Key point: A **probability** (G-1574) can be counted from data (empirical) or from the sample space (theoretical); as the number of trials grows, the empirical value settles on the theoretical one.

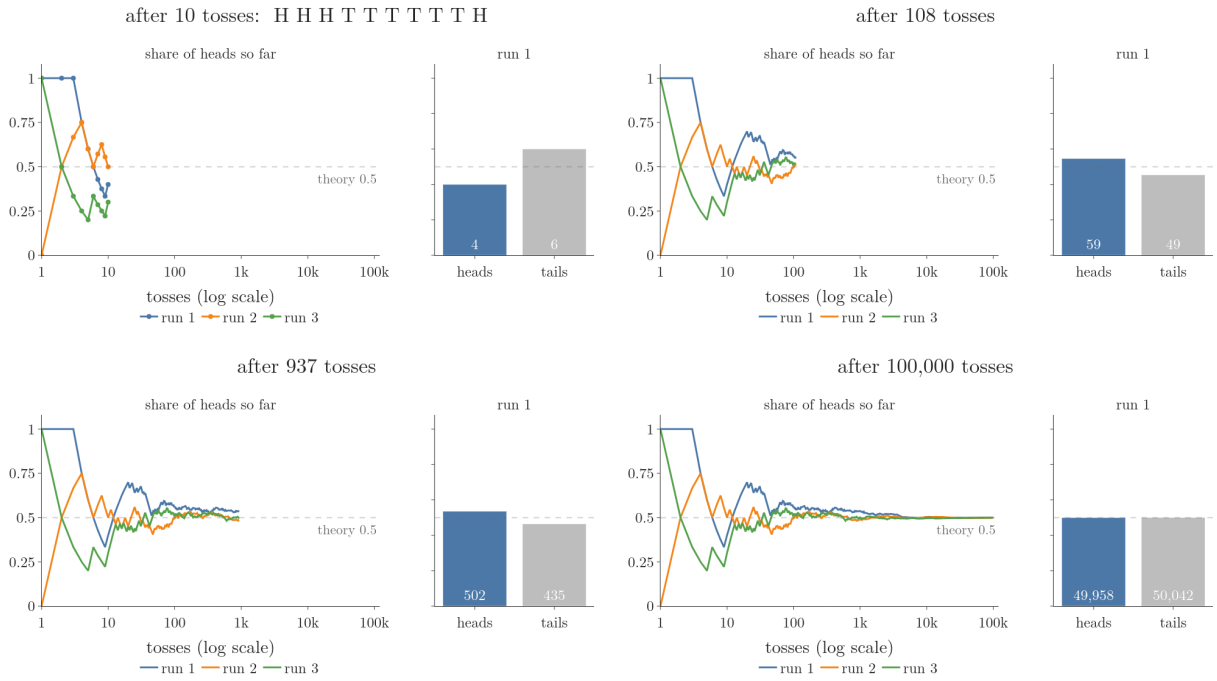


Figure 1: A fair coin tossed up to 100,000 times in three runs: the share of heads so far (left) and run 1's head and tail counts (right) settle on the theoretical 0.5

Figure 1 tosses a fair coin 100,000 times, three separate times, and plots the share of heads after each toss; the bars on the right count run 1's heads and tails (the idea follows Seeing Theory, Kunin et al.). The horizontal axis is on a **log scale**: 1 to 10, 10 to 100, 100 to 1k and so on each take the same width, so the first 10 tosses and the last 90,000 both fit. Watch the three lines: early on the share jumps around between 0 and 1. After thousands of tosses all three runs sit on 0.5, the value we get from the sample space $\{H, T\}$ (the set of the two possible outcomes, written between curly brackets) without tossing at all.

These are the two kinds of probability in this Note. The **empirical probability** (G-678) is

the share we observe in data; the **theoretical probability** (G-1966) is the share we work out from the sample space. A last section adds the rules that every probability obeys.

The terms random experiment, trial, outcome, sample space and event are defined there.

2. What a probability measures

> **Key point:** A probability is a number from 0 to 1 that measures how likely an event is: 0 means it never happens, 1 means it always happens.

A **probability** (G-1574) is a measure of how likely a particular event is to occur. We write the probability of an event A as $P(A)$. A probability always lies between 0 and 1:

- $P(A) = 0$: A never happens (the impossible event).
- $P(A) = 1$: A always happens (the sure event).
- $P(A) = 0.5$: A happens half the time, as heads does on a fair coin.

Figure 2 places a few die and coin events on this scale.

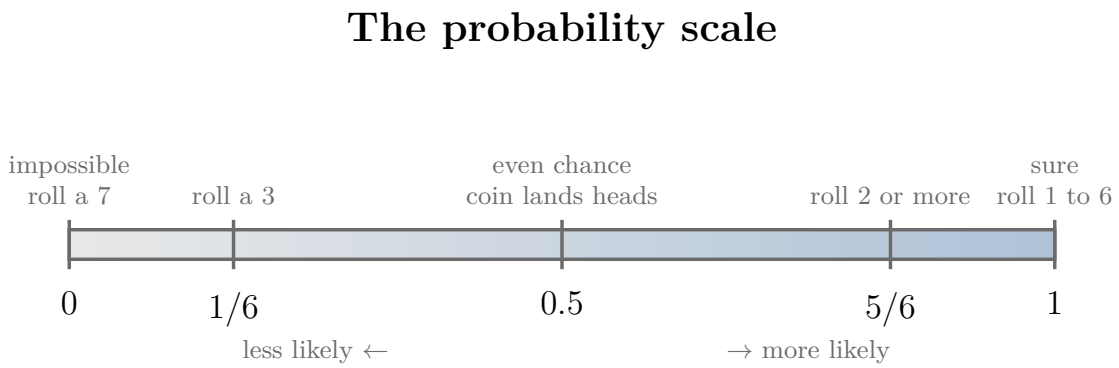


Figure 2: The probability scale from 0 (impossible) to 1 (sure)

Probability is always computed for an event: "getting a head", "rolling a 3", "two heads in two tosses". The next two sections give the two ways of computing it.

3. Empirical probability

> **Key point: Empirical probability** (G-678) is the share of trials in which the event actually happened: count it, then divide by the number of trials.

Empirical probability (also called **experimental probability**) is based on observed data rather than on theoretical assumptions. We run the experiment, or look at data from past runs, and count.

1. **In words:** the number of trials in which the event happened, divided by the total number of trials.

2. **Formula:**

$$P(A) = \frac{\text{times } A \text{ occurred}}{\text{trials}}$$

3. **Example:** a coin tossed 100 times landed heads 55 times and tails 45 times:

$$P(\text{head}) = \frac{55}{100} = 0.55$$

Empirical probability is the **relative frequency** (G-1664; a value's count divided by the total), and the estimated PMF (probability mass function, the **probability** (G-1574) of each value; estimating a PMF), read as the probability of an event.

3.1 A bag of marbles

> **Key point:** Each colour's **empirical probability** (G-678) is its count divided by the 200 draws, whatever the bag holds.

A bag holds 50 marbles: 20 red, 15 blue and 15 green. We draw one marble, note its colour and put it back (**with replacement**, G-2125), 200 times. Red came up 80 times, blue 70 times and green 50 times.

$$P(\text{red}) = \frac{80}{200} = 0.40$$

$$P(\text{blue}) = \frac{70}{200} = 0.35$$

$$P(\text{green}) = \frac{50}{200} = 0.25$$

Only the draws enter the calculation, not the contents of the bag. Section 4 compares these numbers with what the contents predict.

3.2 Titanic passengers

> **Key point:** On a dataset, the **empirical probability** (G-678) of a value is its share of the **observations** (records, one row of the data table each).

The Titanic dataset of drawing a passenger has 891 passengers: 216 in class 1, 184 in class 2 and 491 in class 3. For one passenger drawn at random:

$$P(\text{class 1}) = \frac{216}{891} \approx 0.242$$

$$P(\text{class 2}) = \frac{184}{891} \approx 0.207$$

$$P(\text{class 3}) = \frac{491}{891} \approx 0.551$$

Here there is no formula to fall back on: the data is all we have, so the empirical probability is the **probability** (G-1574). Figure 3 draws the three shares against the "1 out of 3" that a careless count of the sample space would give (the Extra of section 4 explains why that count is wrong).

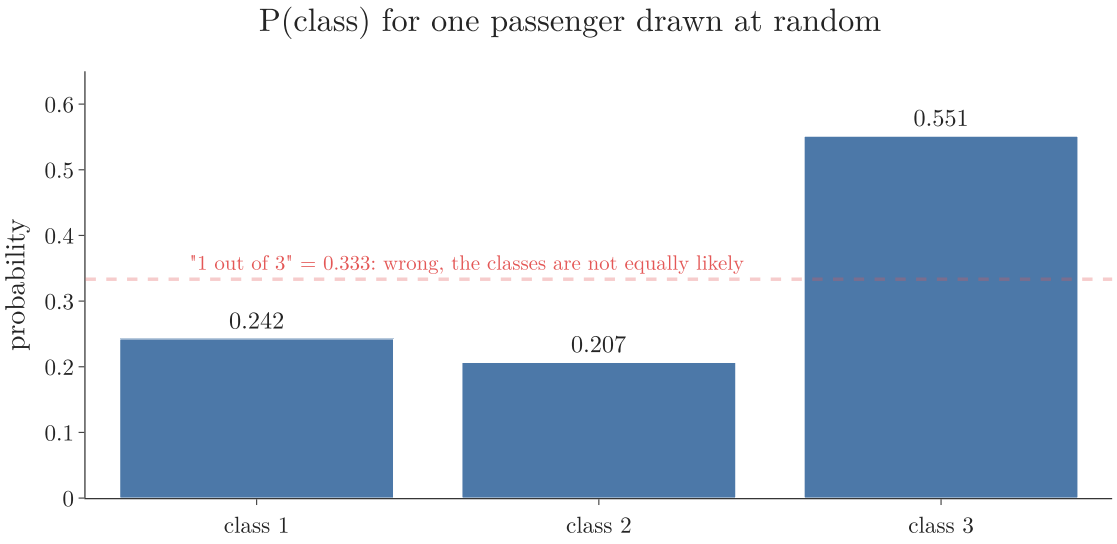


Figure 3: The empirical probability of each Titanic class, from 891 passengers, against the wrong "1 out of 3" (dashed).

> **Python:** The share of each value in a column.
>
> ```
> import pandas as pd
>
> df = pd.read_csv("data/titanic_train.csv")
> # share of rows per class: empirical probabilities
> df["Pclass"].value_counts(normalize=True)
> ```
>
> `normalize=True` divides each count by the number of rows (891). The Notebook (`MA-011-empirical-and-theoretical-probability.ipynb`) runs this and every simulation in this Note.

4. Theoretical probability

> **Key point:** When every outcome is equally likely, the **theoretical probability** (G-1966) of an event is its number of outcomes divided by the size of the sample space; no trials are needed.

Theoretical probability (also called **classical probability**) is worked out from the sample space instead of from data. Theoretical probability applies when each outcome in the sample

space is **equally likely** (G-702), as with a fair die: a 3 is as likely as a 2, a 4 or any other face.

1. **In words:** the number of outcomes in the event (the **favourable outcomes**, G-758) divided by the number of outcomes in the sample space.
2. **Formula:**
$$P(A) = \frac{\text{favourable}}{\text{all outcomes}}$$
3. **Example:** rolling a 3 on a die. Only the outcome 3 is favourable, out of 6:
$$P(3) = \frac{1}{6} \approx 0.167$$

More events, all by counting:

Experiment	Event	Favourable outcomes	Outcomes in S	P
Toss a coin	head	$\{H\}$	2	1/2
Roll a die	a 3	$\{3\}$	6	1/6
Roll a die	less than 3	$\{1, 2\}$	6	$2/6 = 1/3$
Toss a coin twice	two heads	$\{HH\}$	4	1/4
Toss a coin twice	at least one head	$\{HH, HT, TH\}$	4	3/4

No trial was run for any of these. The **probability** (G-1574) comes from the structure of the experiment alone.

For the marble bag, each of the 50 marbles is equally likely to be drawn, so the theory gives:

$$P(\text{red}) = 20/50 = 0.4$$

$$P(\text{blue}) = 15/50 = 0.3$$

$$P(\text{green}) = 15/50 = 0.3$$

The 200 draws of section 3.1 gave:

Colour	Empirical (200 draws)	Theoretical (bag contents)
Red	$80/200 = 0.40$	$20/50 = 0.40$
Blue	$70/200 = 0.35$	$15/50 = 0.30$
Green	$50/200 = 0.25$	$15/50 = 0.30$

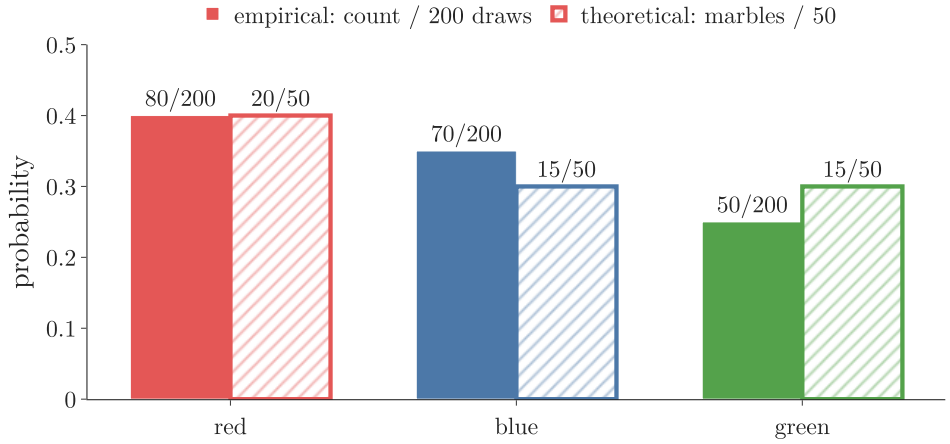


Figure 4: The marble bag: empirical shares from 200 draws (solid) against the theoretical shares from the bag's contents (hatched)

Figure 4 puts the two side by side. Red matches exactly; blue and green are 0.05 away. With only 200 draws, a gap of this size is normal. The count of blue marbles in 200 draws is binomial (a count of successes in repeated trials), so the share of blue has this standard deviation (a typical size of the gap), with $p = 0.3$ the true share and $n = 200$ the draws:

$$\sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{0.3 \times 0.7}{200}} = \sqrt{0.00105} = 0.032$$

The gap in these units:

$$\frac{0.05}{0.032} = 1.56$$

A gap of 0.05 is about 1.5 of these standard deviations, well within the usual spread. Section 5 shows the gaps shrinking as the trials grow.

Extra: The formula only holds when the outcomes are equally likely, and getting this wrong is a common mistake:

- **Titanic classes:** the sample space $\{1, 2, 3\}$ has three outcomes, but "1 out of 3" gives $P(\text{class } 3) = 0.333$, while the data says 0.551. Classes are not equally likely, so only the empirical value is right.
- **Sum of two dice:** the sums $2, \ldots, 12$ are 11 outcomes, but $P(\text{sum } 7)$ is not $1/11$. The equally likely outcomes are the 36 pairs, and 6 of them give 7 (see the sum of two dice):

$$P(\text{sum } 7) = 6/36$$

The fix is to write the sample space as outcomes that really are equally likely (the 36 pairs), and count those.

5. From empirical to theoretical

> **Key point:** The more trials we run, the closer the **empirical probability** (G-678) gets to the theoretical one; with few trials they can be far apart.

The empirical and theoretical probabilities of the same event are rarely equal after a few trials. As the number of trials grows, the empirical value closes in on the theoretical one. This settling is the **law of large numbers** (G-1052), met in estimating a PMF by simulation.

Ten tosses of a fair coin can easily give 3 heads, an empirical probability of 0.3 against the theoretical 0.5. One simulated coin, tossed more and more times:

Tosses	10	100	1,000	10,000	100,000
Empirical $P(\text{head})$	0.4	0.55	0.506	0.4918	0.5018
Distance from 0.5	0.1	0.05	0.006	0.0082	0.0018

The distance does not shrink at every step (10,000 tosses landed a little further away than 1,000), but the overall trend is clear.

The typical-gap formula of section 4 shows why the trend holds: the number of trials n sits under the square root, in the bottom of the fraction. For a fair coin, $p = 0.5$:

$$n = 100: \quad \sqrt{\frac{0.5 \times 0.5}{100}} = 0.05$$

$$n = 10000: \quad \sqrt{\frac{0.5 \times 0.5}{10000}} = 0.005$$

So 100 times more tosses make the typical gap 10 times smaller.

Figure 1 shows the same pattern toss by toss in three runs: wild swings at first, a narrow band after 1,000 tosses, and a flat line at 0.5 near 100,000.

A die behaves the same way. In 10 simulated rolls, the face 3 did not appear at all, an empirical $P(3) = 0$. In 100,000 rolls it appeared in a share of 0.1662, against the theoretical $1/6 \approx 0.1667$. Figure 5 shows all six faces at 10, 1,000 and 100,000 rolls: the bars start ragged and end level with the dashed line.

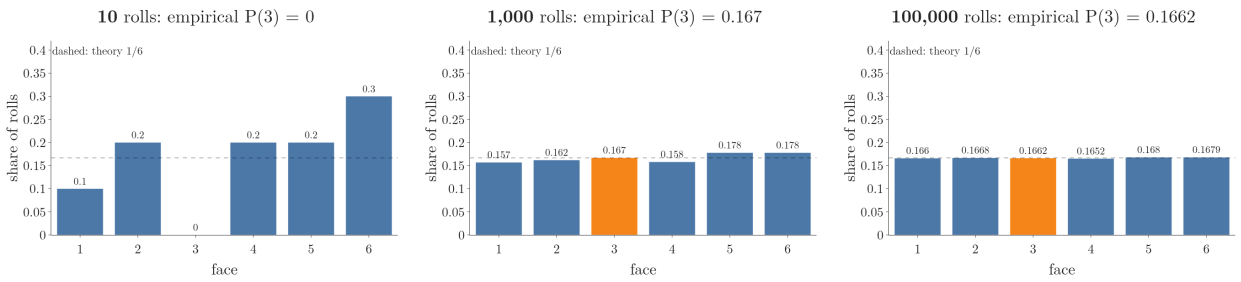


Figure 5: The share of each face in 10, 1,000 and 100,000 simulated rolls of a die (the Notebook's draws), against the theoretical 1/6 (dashed). Face 3 is orange.

Python: Tossing a coin many times.

```
import numpy as np

rng = np.random.default_rng(42)        # fixed seed
for n in [10, 100, 1_000, 10_000, 100_000]:
    tosses = rng.integers(0, 2, n)     # 0 = tail, 1 = head
    print(n, tosses.mean())
```

The mean of 0s and 1s is the share of 1s, so `tosses.mean()` is the empirical probability of a head. The seed makes every run give the numbers in the table above.

5.1 Which one to use

Key point: Theoretical probability (G-1966) needs a known sample space of equally likely outcomes; empirical probability needs only data.

- **Theoretical:** when we know the sample space and its outcomes are **equally likely** (G-702) (coins, dice, cards). Theoretical probability shows how the event behaves in principle, before any data exists.
- **Empirical:** when the outcomes are not equally likely or the sample space is not fully known, which is the usual case with real data. Statistics and hypothesis testing work with empirical probabilities.

Machine learning models learn from data, and many of their probabilities are counted from it. The class priors of Naive Bayes (the share of each class), such as 5 wins in 8 matches, are counted from the training observations.

6. The rules every probability follows

Key point: Three rules, the axioms, define every **probability** (G-1574); from them follow the complement rule $P(\text{not } A) = 1 - P(A)$ and the addition rule $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

Extra: This whole section is added material: the basic rules that later Notes use without proof.

6.1 The axioms of probability

> **Key point:** Probabilities are never negative, the whole sample space has **probability** (G-1574) 1, and the probabilities of events with no shared outcomes add up.

Every probability, empirical or theoretical, obeys three rules called the **axioms of probability** (G-240). Everything else about probability is derived from them. Modern probability rests on axioms of this kind, due to Kolmogorov (MML §6.1.2); the three below, and the **complement** (G-422) and general addition rules of sections 6.2 and 6.3, are Theorems 1.1 and 1.4 of Grinstead and Snell (1997, §1.2).

1. **Non-negative:** $P(A) \geq 0$ for every event A.
2. **The sure event has probability 1:** $P(S) = 1$.
3. **Mutually exclusive events add:** if A and B share no outcome, $P(A \cup B) = P(A) + P(B)$, where $A \cup B$ is the event that A happens, or B happens, or both. Axiom 3 is the addition rule for mutually exclusive events.

The sign $\varnothing$ is the **empty event**, the event with no outcomes (for a die, "a 7"). Two facts follow.

The empty event has probability 0. S and $\varnothing$ share no outcome, and joining them gives S back. So axiom 3 gives:

$$P(S) = P(S) + P(\varnothing)$$

Taking $P(S)$ away from both sides:

$$P(\varnothing) = 0$$

No probability is above 1. The complement rule of section 6.2 gives $P(A) = 1 - P(A^c)$, and axiom 1 says $P(A^c) \geq 0$. Taking away a number that is not negative cannot go above 1:

$$P(A) = 1 - P(A^c) \leq 1$$

Together these give the 0-to-1 scale of Figure 2.

6.2 The complement rule

> **Key point:** The **probability** (G-1574) that A does not happen is 1 minus the probability that it does.

The **complement** (G-422) of an event A, written A^c (or $\bar{A}$, "not A"), is the event made of every outcome in the sample space that is not in A. On a die, the complement of "odd" $= \{1, 3, 5\}$ is "even" $= \{2, 4, 6\}$.

A and A^c are mutually exclusive and exhaustive: exactly one of them happens in every trial. So by axioms 2 and 3, $P(A) + P(A^c) = 1$.

1. **In words:** the chance that A does not happen is whatever is left of 1 after the chance that it does.
2. **Formula:**
$$P(A^c) = 1 - P(A)$$
3. **Example:** the chance of at least one head in two tosses. Its complement is "no heads" $= \{TT\}$, with probability 1/4:
$$P(\text{at least one head}) = 1 - P(TT) = 1 - \frac{1}{4} = \frac{3}{4}$$

The result matches the count in section 4. The complement is often the shortcut: "at least one" events have many outcomes, while their complement "none" has one. Figure 6 shows why the shortcut works: the four outcomes split into the three in A and the one in its complement.

$$\mathrm{P(A)} = 1 - \mathrm{P(A^c)} = 1 - 1/4 = 3/4$$

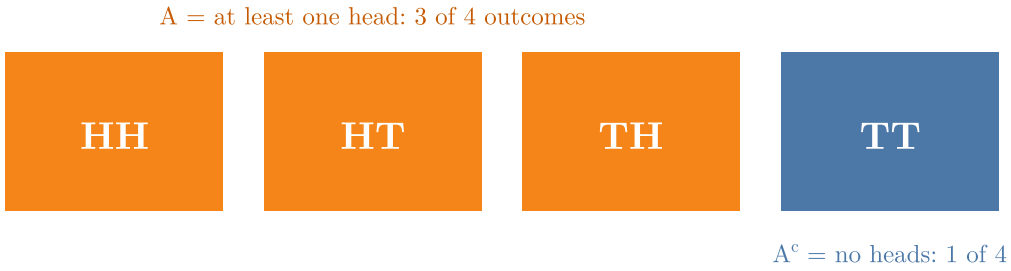


Figure 6: The two-toss sample space split into A = at least one head (orange, 3 outcomes) and its complement, no heads (blue, 1 outcome).

6.3 The general addition rule

> **Key point:** For events that can happen together, adding their probabilities counts the shared outcomes twice, so we subtract the overlap once.

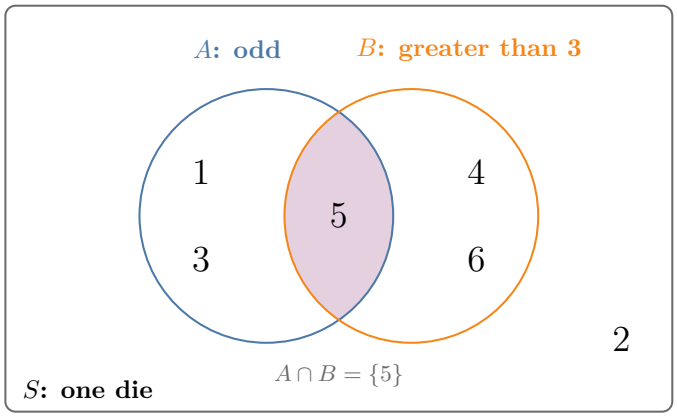
The addition rule of axiom 3 holds only for mutually exclusive events. When A and B share outcomes, $P(A) + P(B)$ is too large.

Take one die, with $A =$ "odd" $= \{1, 3, 5\}$ and $B =$ "greater than 3" $= \{4, 5, 6\}$. Figure 7 is a Venn diagram: the box is the sample space, each circle holds the outcomes of one event, and an outcome inside both circles belongs to both events. It shows that 5 belongs to both, so adding the two probabilities counts it twice:

$$P(A) + P(B) = 3/6 + 3/6 = 1$$

Yet $A \cup B = \{1, 3, 4, 5, 6\}$ misses 2, so its **probability** (G-1574) must be below 1. Two signs name the parts: $A \cup B$ (read: A or B) is every outcome in at least one of the two events, and $A \cap B$ (read: A and B) is the overlap, the outcomes in both. Here $A \cap B = \{5\}$.

Adding $P(A)$ and $P(B)$ counts the overlap twice



$P(A) = 3/6, \;\; P(B) = 3/6$

$3/6 + 3/6 = 6/6$: too much, 5 counted twice

$P(A \cup B) = 3/6 + 3/6 - 1/6 = 5/6$

Figure 7: The general addition rule on one die: the overlap $\{5\}$ is counted twice by $P(A) + P(B)$

1. **In words:** add the two probabilities, then subtract the probability of the overlap, which was counted twice.
2. **Formula:**
$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$
3. **Example:** with $A \cap B = \{5\}$,
$$P(A \cup B) = \frac{3}{6} + \frac{3}{6} - \frac{1}{6} = \frac{5}{6}$$
Counting directly: $A \cup B = \{1, 3, 4, 5, 6\}$ has 5 of the 6 outcomes, so 5/6.

For mutually exclusive events $P(A \cap B) = 0$, and the general rule shrinks back to axiom 3.

7. Summary

	Empirical probability	Theoretical probability
Also called	Experimental	Classical
Computed from	Observed trials or data	The sample space
Formula	times A occurred / trials	favourable outcomes / all outcomes
Needs	Data	Equally likely outcomes
Coin, head	$55/100 = 0.55$	$1/2$
Changes with more data?	Yes, settles on the theoretical value	No

- Use **theoretical probability** (G-1966) when the sample space is known and its outcomes are equally likely (coins, dice, cards); use empirical probability otherwise, which is the usual case with real data.

Rule	Formula	Die example	Why it matters
Complement	$P(A^c) = 1 - P(A)$	$P(\text{not } 6) = 5/6$	an "at least one" event is often easier through its complement "none"
Addition, mutually exclusive	$P(A \cup B) = P(A) + P(B)$	$P(1 \text{ or } 2) = 2/6$	no shared outcome, so nothing is counted twice
Addition, general	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$	$P(\text{odd or} > 3) = 5/6$	adding counts the overlap twice, so it is subtracted once

- A **probability** (G-1574) lies between 0 (impossible) and 1 (sure), because the axioms allow no negative probability and give the whole sample space probability 1.
- Theoretical probability fails when the outcomes are not **equally likely** (G-702); then only data can tell.
- With many trials, the **empirical probability** (G-678) approaches the theoretical one; with few, it can be far off, so trust a probability counted from data more when it comes from many trials.
- Probabilities such as Naive Bayes class priors are estimated empirically, from the training data, because real data rarely has equally likely outcomes.

8. Sources

Built from

- CampusX, "Master Probability in Data Science: The Ultimate Crash Course! | Part 1 | CampusX", YouTube, https://www.youtube.com/watch?v=DUT4WEUngt0

Other references

- Deisenroth, M. P., Faisal, A. A. and Ong, C. S. (2020). *Mathematics for Machine Learning* (MML). Cambridge University Press. §6.1.2.
- Grinstead, C. M. and Snell, J. L. (1997). *Introduction to Probability*, 2nd ed. American Mathematical Society. §1.2, Theorems 1.1 and 1.4.
- Kunin, D., Guo, J., Devlin, T. D. and Xiang, D. *Seeing Theory*, Brown University, chapter 1 "Basic Probability", section "Chance Events". seeing-theory.brown.edu/basic-probability. The idea of Figure 1, watching the share of heads settle as the tosses grow, follows its coin-flipping demonstration; data and code are our own.

9. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Probability (G-1574)	A number from 0 to 1 measuring how likely an event is; written $P(A)$.
Empirical (experimental) probability (G-678)	The share of trials in which an event happened, counted from data. It estimates the probability when it cannot be worked out from the sample space.
Theoretical (classical) probability (G-1966)	Favourable outcomes divided by all outcomes, for equally likely outcomes.
Equally likely outcomes (G-702)	Outcomes that all have the same probability, such as the faces of a fair die.
Favourable outcome (G-758)	An outcome that belongs to the event we are measuring.
Axioms of probability (G-240)	The three rules every probability obeys: non-negative, $P(S) = 1$, mutually exclusive events add.
Complement (A^c) (G-422)	The event that A does not happen: every outcome not in A.
Complement rule (G-423)	The probability that an event does not happen is 1 minus the probability that it does, $P(A^c) = 1 - P(A)$; useful when the opposite event is easier to count, as in at least one head.
Relative frequency (G-1664)	A category's share of all the data: its frequency divided by the total.
With replacement (G-2125)	Sampling in which each drawn item is put back, so it can be drawn again.
Law of large numbers (G-1052)	The more trials, the closer a share of trials gets to the true probability.

Term	Meaning
Addition rule (general addition rule) (G-173)	The rule for the probability that A or B (or both) happens: add their probabilities and subtract the overlap so it is not counted twice, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$; for mutually exclusive events the overlap is 0.